Major Project Stage-1 Review-2 Presentation
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Predictive Thermal Battery Management System for Li-ion Battery Packs

Presented by

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Abstract

- Lithium-ion cells offer high energy density but suffer catastrophic thermal runaway under internal defects, dendrite formation, overcharging, and soft micro-shorts.
- Conventional BMS boards rely solely on fixed passive thresholds (e.g., 55 °C surface limit). They act reactively after the outer can is already hot, cannot detect uncooled core heat, and are completely blind to faults at zero current.
- This project develops an autonomous, predictive, dual-layer battery management system for a 4S1P Panasonic NCR18650GA Li-ion pack:
- • Layer A (Hardware BMS): Autonomous, non-programmable backstop for primary short-circuit and cell over/under-voltage protection.
- • Layer B (Predictive Supervisor): ESP32 MCU running a 2-node lumped-parameter thermal observer and discrete Kalman filter at 5 Hz.
- Key Engineering Advantages:
- • Physics-based estimation running directly on a low-cost $4 microcontroller without GPU/cloud dependency.
- • Detects abnormal heat generation (Q_ab in Watts) down to 0.2 W, isolating core faults before external heating.
- • Provides 5-minute thermal forecasting and autonomous relay contactor disconnection, acting 5.0 minutes earlier and keeping cells 27.1 °C cooler.

Problem Statement

- Thermal runaway in Li-ion cells initiates inside the electrode jellyroll via internal defects, dendrites, or local micro-shorts before surface temperature rises.
- Critical Vulnerabilities of Existing Battery Management Systems:
- 1. Fixed Reactive Thresholds: Intervene only after surface exceeds 55 °C or 60 °C—by then, internal exothermic chain reactions are already self-sustaining.
- 2. Core Thermal Blindspot: Surface NTCs lag core temperature by several degrees during dynamic discharge; internal runaway begins at the core.
- 3. Zero-Current Rest Blindspot: When pack current is zero (vehicle parked / charging paused), conventional BMS cannot detect internal leakage heating.
- 4. Deep-Learning Limitations: Research algorithms (like our base paper) use heavy LSTM networks requiring GPU training, retraining as cells age, and cannot run on cheap microcontrollers.
- Need: An on-board, embedded, physics-based observer that isolates abnormal fault heat in Watts and acts preventatively before runaway becomes unstoppable.

Literature Survey

S.No	Paper Title	Outcomes	Limitation / Research Gap
[1]	Thermal runaway mechanism of lithium ion battery for electric vehicles: A review (Feng et al., 2018)	Reviewed how thermal runaway starts and spreads through a chain of heat-producing reactions inside the cell.	Explains reaction physics, but does not give an on-board method to detect it early.
[2]	Battery management system: An overview of its application in the smart grid and electric vehicles (Rahimi-Eichi et al., 2013)	Showed that a BMS needs accurate state estimation and ways to protect the battery from hazardous conditions.	Focuses on state-of-charge and state-of-health estimation, not on early thermal warning.
[3]	Protection devices in commercial 18650 lithium-ion batteries (Xu et al., 2021)	Opened four commercial 18650 cells. Only current interrupt device (CID) and top vent are fitted in every cell.	These are destructive last-resort devices. Pack-level monitoring and early warning are not covered.
[4]	Advanced fault diagnosis for lithium-ion battery systems: A review of fault mechanisms (Hu et al., 2020)	Reviewed internal battery faults, sensor faults and actuator faults, and how each one can be diagnosed.	A comprehensive review only. It does not build or test a low-cost embedded diagnostic system.
[5]	A critical review of thermal runaway prediction and early-warning methods (Zhang et al., 2023)	Grouped early-warning methods into electrochemistry-based, big-data and artificial intelligence methods.	Existing methods rely on complex electrochemical models, massive datasets, or heavy AI training.

Literature Survey (contd.)

S.No	Paper Title	Outcomes	Limitation / Research Gap
[6]	An online data-driven fault diagnosis and thermal runaway early warning for electric vehicle batteries (Sun et al., 2022)	Found the cell heading for thermal runaway before it happened, using voltage and temperature data from real vehicles.	Compares many cells statistically in large EV packs; unsuitable for standalone small packs.
[7]*	BASE PAPER: A neural network based method for thermal fault detection in lithium-ion batteries (Ojo et al., IEEE TIE, 2021)	LSTM neural network predicts surface temperature 4.5 s ahead; flags gaps > 0.69 °C as thermal runaway precursor.	Requires 65 min GPU training, retraining as cell ages; blind to slow faults (< 9.2 °C/min); 0.5% false alarms (~36/hr).
[8]	A combined data-driven and model-based algorithm for accurate battery thermal runaway warning (Chen et al., 2024)	Combined K-Means clustering with the Bernardi heat equation and warned 25 minutes before thermal runaway.	Its 1 °C/s limit confirms runaway after initiation; it is not an early preventative warning.
[9]	Online parameterization of lumped thermal dynamics in cylindrical lithium ion batteries (Lin et al., 2013)	Showed that only surface temperature can be measured while core is hotter, and estimated core temperature.	Needs complex online parameter identification, and was tested on one 26650 cell, not a 4S pack.
[10]	Cloud-based battery condition monitoring platform for large-scale BESS using IoT (Adhikaree et al., 2017)	Sent battery module data over IoT links to Google Cloud, where cell states were evaluated.	Depends strictly on the cloud and network. Protection must keep working even when Wi-Fi is lost.

* Designated Base Paper of this project (IEEE Trans. Ind. Electron., 2021)

Problem Analysis

- Internal heat generation ($I^2R + Q_{ab}$) drives thermal runaway inside the electrode assembly.
- Surface temperature lags core temperature by several minutes due to internal thermal resistance ($R_{core} = 2.0$ °C/W).
- Fixed threshold protection (55 °C) provides zero early-warning time; once exothermic reactions start, disconnecting current cannot quench runaway.
- Incipient internal micro-shorts generate very small heat powers (0.03 W to 0.29 W) that take tens of minutes to warm the cell, easily missed by short-window detectors.

Research Gap

- Need for a lightweight, physics-based thermal observer executable on a low-cost microcontroller (ESP32) at 5 Hz without GPU/cloud dependency.
- Need for abnormal heat (Q_{ab}) isolation in physical Watts to detect micro-faults even when the pack is at zero-current rest.
- Need for dual-layer safety: an independent, non-programmable hardware BMS backed by a predictive software supervisor with separate charge/load disconnects.

Objectives of the Project

- 1. To formulate and validate a lumped-parameter two-node electro-thermal model capturing core and surface dynamics of 18650 Li-ion cells.
- 2. To design an autonomous dual-layer safety architecture: an independent hardware BMS (Layer A) backed by an ESP32 supervisor (Layer B) with directional load and charge disconnects.
- 3. To implement high-resolution analog front-end sensing: 16-bit individual cell voltages (ADS1115), pack current (ACS712 / INA226), and five NTC thermistors.
- 4. To formulate and execute a real-time discrete Kalman filter at 5 Hz to estimate core temperature (T_core) and isolate abnormal fault heat (Q_ab) in Watts, latching trips in flash.
- 5. To validate the system through KiCad 10 circuit simulations, automated safety testbenches, and prototype hardware bring-up.

Methodology & Safety Algorithm Flowchart

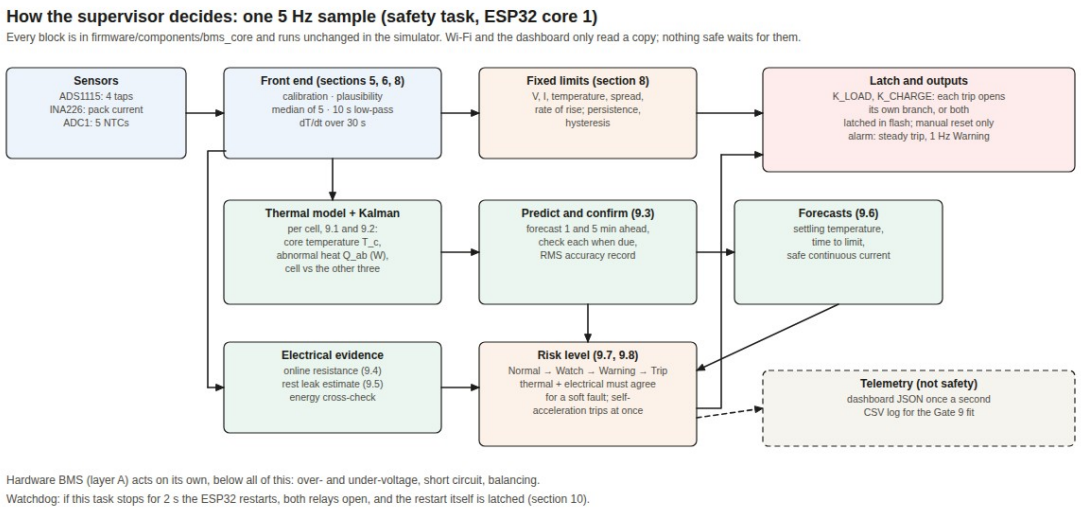
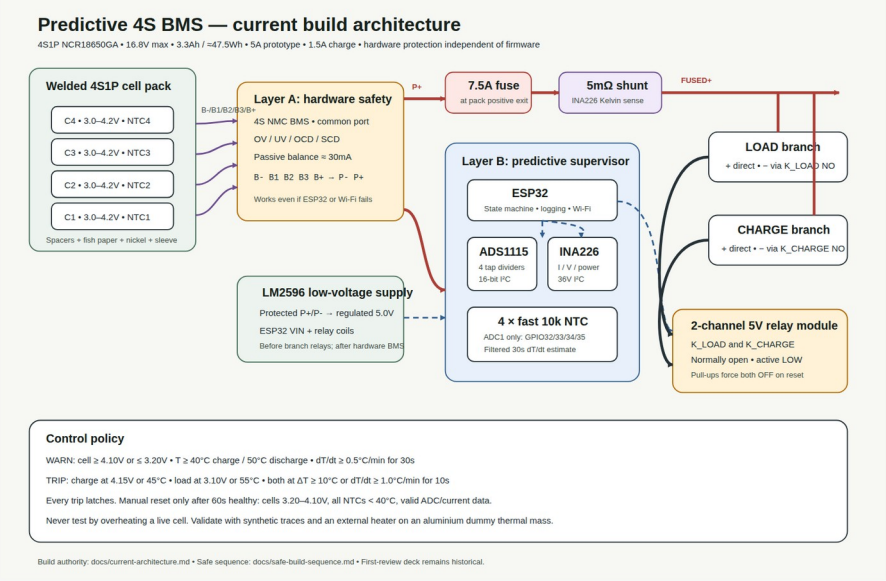


Figure: FreeRTOS 5 Hz deterministic safety pipeline and graduated protection state machine

- Deterministic 5 Hz Safety Cycle: FreeRTOS safety task executes precisely every 200 ms.
- 3-Stage Signal Conditioning: 5-sample median filter rejects spikes; 10 s LPF removes noise; 30 s regression computes dT/dt.
- Kalman State Observer: Estimates core temperature and isolates abnormal defect heat Q_ab in Watts.
- Graduated Safety Actuation: Normal -> Watch -> Warning (1 Hz buzzer) -> Latched Trip (relays open, logged to flash).

System Architecture & Block Diagram

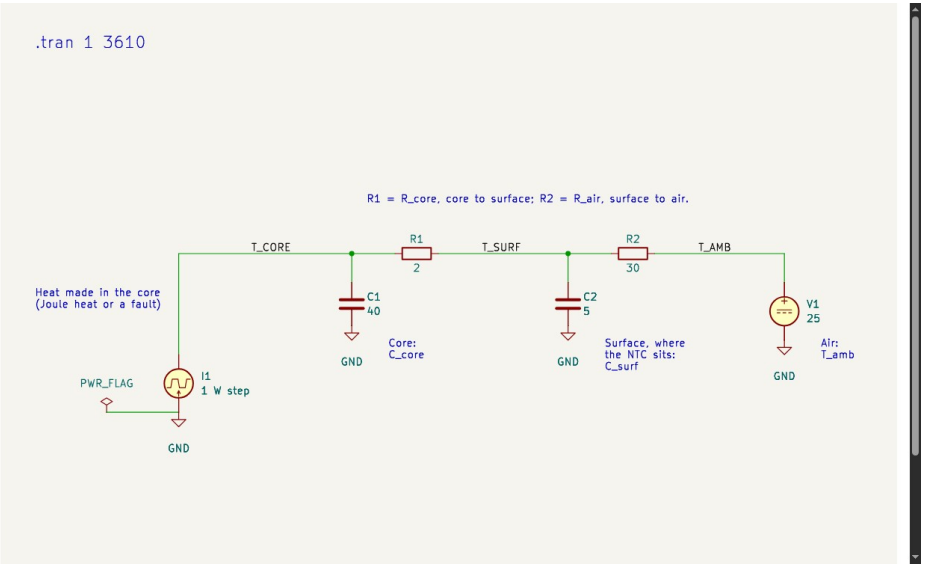


- Welded 4S1P Cell Pack: Panasonic NCR18650GA (14.4 V nom, 16.8 V max, 3.3 Ah / 47.5 Wh) with cell NTC thermistors.
- Layer A (Hardware Safety BMS): Direct cell connection providing non-programmable OV (4.25 V), UV (2.70 V), OCD, SCD, and passive balancing.
- Primary Pack Fuse: 7.5 A fast blade fuse directly at positive exit terminal for ultimate short-circuit protection.
- Board A (Cell Taps): Conditions cell voltages via 330k/56k dividers and 10k/100nF RC filters into ADS1115 16-bit ADC.
- Board B (Supervisor): ESP32 running 5 Hz Kalman observer; monitors current via ACS712 / INA226 Kelvin shunt.
- Fail-Safe Contactor Disconnects: Normally-open K_LOAD and K_CHARGE relays in negative return (P-); hardware pull-ups ensure open on power loss.
- Isolated Auxiliary Power: LM2596 buck converter (5.0 V) tapped before current sensor so MCU draw is excluded from rest monitoring.

Mathematical Formulation: Thermal Model

- A lumped-parameter two-node electro-thermal circuit models core and surface dynamics:
- Core Node Differential Equation:
- $C_{core} \cdot (d T_{core} / dt) = I^2 R_{int} + Q_{ab} - (T_{core} - T_{surf}) / R_{core}$
- Surface Node Differential Equation:
- $C_{surf} \cdot (d T_{surf} / dt) = (T_{core} - T_{surf}) / R_{core} - (T_{surf} - T_{amb}) / R_{air}$
- Physical Model Parameters (for 18650 3.3 Ah cell):
- • C_{core} = 40.0 J/°C, C_{surf} = 5.0 J/°C (Thermal capacities)
- • R_{core} = 2.0 °C/W (Internal conduction resistance from core to can)
- • R_{air} = 30.0 °C/W (Convection resistance from can surface to ambient air)
- • $I^2 R_{int}$: Known electrical Joule heating ($R_{int} \approx 35$ mΩ nominal)
- • Q_{ab}: Unknown abnormal fault heat generation in Watts (internal micro-short, defect).
- Hierarchical Protection Action:
- • Level 0 (Normal): $Q_{ab} < 0.25$ W | Level 1 (Watch): $0.25 \text{ W} \le Q_{ab} < 0.38$ W
- • Level 2 (Warning): $Q_{ab} \ge 0.38$ W for 30 s | Level 3 (Trip): $Q_{ab} \ge 0.75$ W for 10 s or $T_{core} \ge 60$ °C

KiCad Circuit Simulation Model



- Thermal Model Mapped to KiCad Circuit Domain: 1 V = 1 °C, 1 A = 1 W, 1 F = 1 J/°C, 1 Ω = 1 °C/W.
- Core Thermal Capacitor: C_core = 40.0 F, representing internal heat capacity of cell jellyroll.
- Surface Thermal Capacitor: C_surf = 5.0 F, representing steel can thermal capacity.
- Conduction Resistor: R_core = 2.0 Ω, modeling internal core-to-surface heat conduction.
- Convection Resistor: R_air = 30.0 Ω, modeling natural convection from can to ambient air.
- Transient Stimulus: 1.0 W pulsed heat injection (current source I1) in cell core from t = 10 s.
- SPICE Simulation Setup: KiCad 10 / ngspice .tran 1 3600 analysis validates transient and steady-state thermal behavior.

Figure: KiCad 10 schematic of the two-node electro-thermal equivalent circuit

KiCad Transient Simulation Results

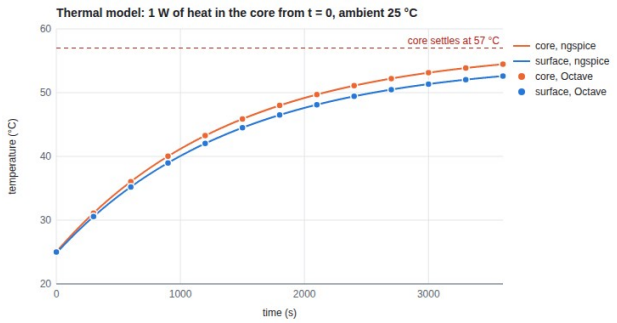


Figure: KiCad / ngspice transient thermal step response: Core (red) and Surface (green) waveforms

- Transient Temperature Response: 1.0 W heat step injected at t = 0 with ambient temperature at 25.0 °C.
- Core Temperature Dynamics: Core temperature rises immediately to 27.0 °C due to internal thermal resistance R_core (2.0 °C/W).
- Surface Temperature Dynamics: Surface temperature lags by several minutes, governed by thermal conduction and convection delay.
- Steady-State Verification: Surface settles at $\Delta T = Q \times R_{air} = 1\ W \times 30\ °C/W = 30\ °C$ above ambient (55.0 °C total).
- Core Steady-State: Core settles at $\Delta T = Q \times (R_{core} + R_{air}) = 1\ W \times 32\ °C/W = 32\ °C$ above ambient (57.0 °C total).
- Thermal Time Constant: Measured $\tau = 1350$ s (22.5 min), confirming theoretical lumped capacitance calculations.

Dynamic Defect Simulation & 5-Min Forecast

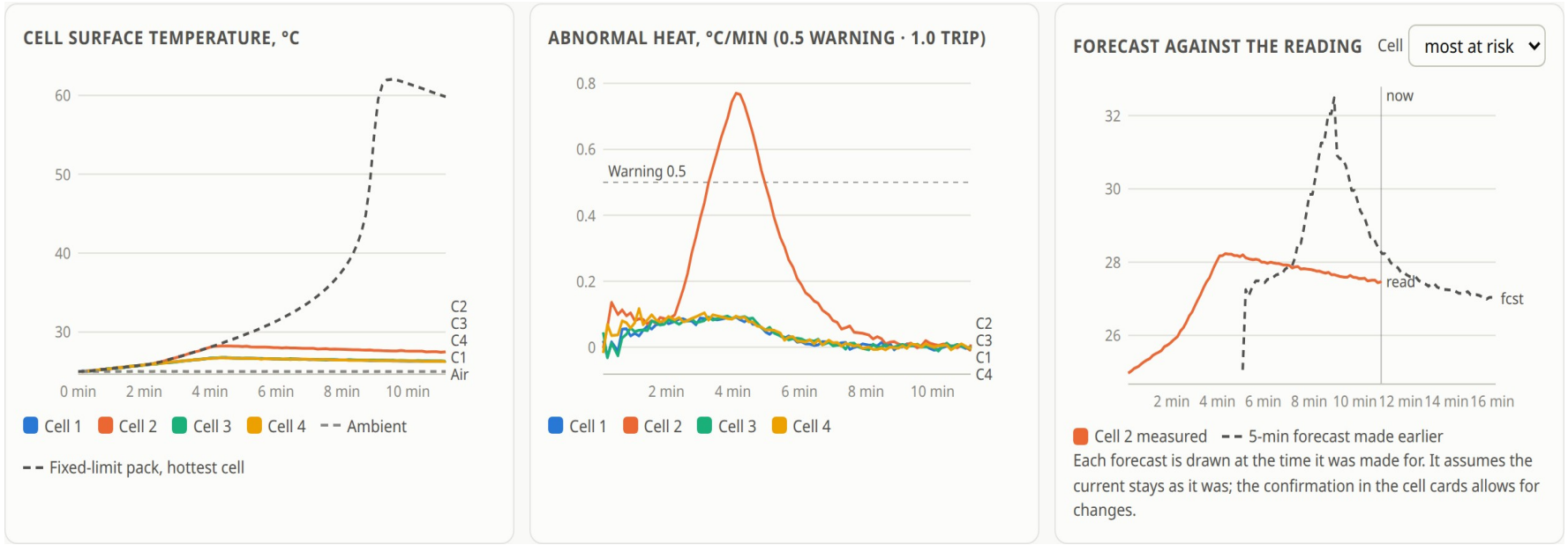


Figure: Defect simulation telemetry: Cell 2 core overheating, Q_ab rising to 1.5 W, and predictive contactor trip

- Fault Injection Scenario: 3.0 A pack discharge combined with a 1.5 W internal defect injected into Cell 2.
- Core Divergence: Cell 2 core climbs rapidly to 31.8 °C while surface lags at 28.1 °C.
- Predictive Trip: Observer detects Q_ab crossing 0.75 W; triggers Level 3 Trip at t = 112 s (surface at only 28.1 °C).
- 5-Minute Forecast Accuracy: RMS forecast error across cells 1–4 is 0.11 °C to 0.12 °C (well within < 0.5 °C qualification limit).

Performance Comparison: Predictive vs Fixed BMS

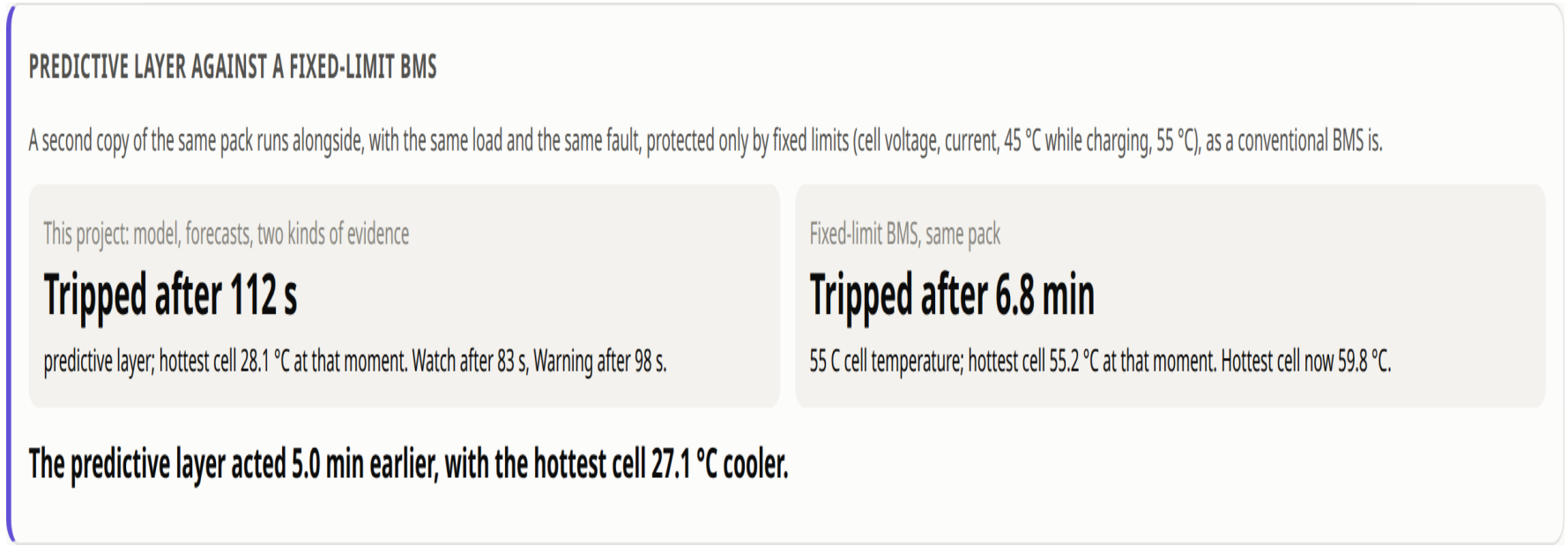
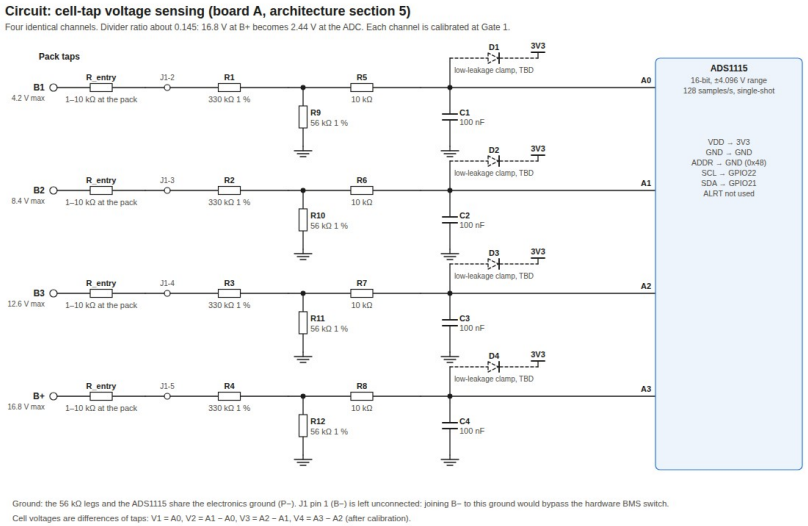


Figure: Side-by-side run comparison: Predictive BMS (trip at 112 s, 28.1 °C) vs Fixed-Limit BMS (trip at 408 s, 55.2 °C)

- Intervention Timing: Predictive BMS trips contactors at 112 s vs 408 s (6.8 min) for conventional BMS (acts 5.0 min earlier!).
- Thermal Suppression: Hottest cell reaches only 28.1 °C under predictive protection vs 55.2 °C for conventional BMS (27.1 °C cooler!).
- Mechanism: Predictive observer acts on abnormal heat rate (Watts), whereas conventional BMS waits for outer can to reach 55 °C.
- Zero-Current Rest Advantage: Detects 1.0 W micro-short in 58 seconds even when current is zero (conventional BMS is completely blind).

Hardware Implementation: Board A (Cell Taps)



- Function: Conditions and filters four series cell tap voltages (B1, B2, B3, B+).
- Precision Resistor Dividers: Four 330 kΩ over 56 kΩ 1% metal-film networks attenuate 16.8 V pack to 2.44 V for ADC range.
- Low-Pass RC Filtering: 10 kΩ series resistor + 100 nF ceramic capacitor suppress switching noise (fc = 27.5 Hz).
- ADS1115 ADC: 16-bit delta-sigma converter over I2C (0x48) provides 125 µV LSB resolution for precision rest-leak tracking.
- Safety Grounding Isolation: J1 pin 1 (B-) left unconnected; prevents thin sense wire from carrying high pack load current.
- Assembly Status: Board A perfboard fully assembled, soldered, and bench verified across all 4 channels.

Figure: Board A cell-tap conditioning schematic with 330k/56k dividers and ADS1115 ADC

Hardware Implementation: Board B (Supervisor Bring-Up)

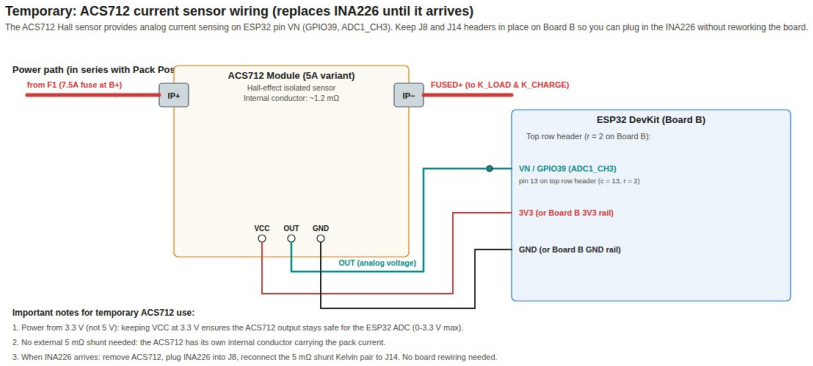
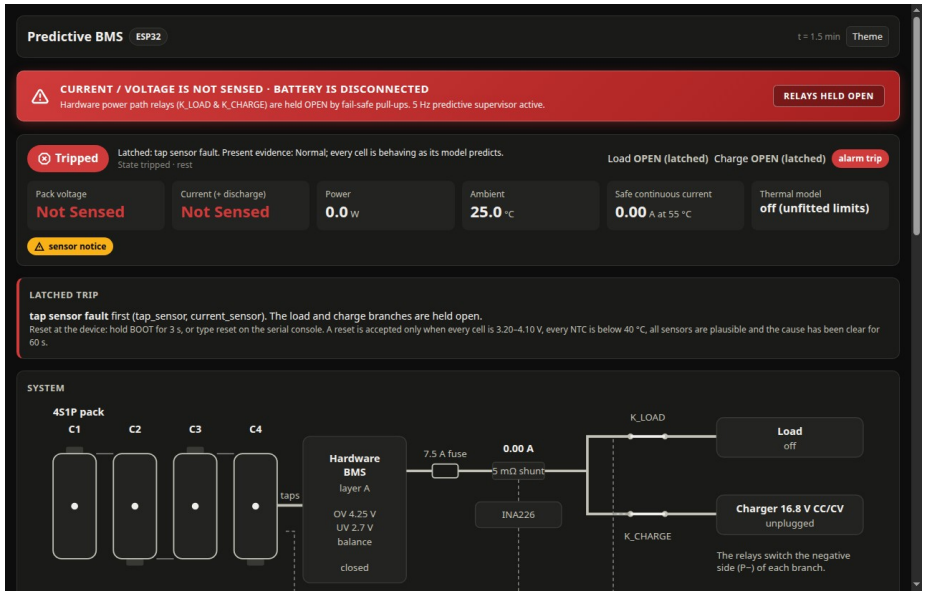


Figure: Board B supervisor architecture with ACS712 current sensing fallback

- Microcontroller: ESP32-WROOM-32 DevKit running FreeRTOS with ADC1 dedicated to five NTC thermistors (10k B3950).
- Current Sensing Fallback: ACS712-5A Hall-effect module wired on ADC1_CH3 (GPIO39/VN); firmware auto-detects and runs telemetry.
- Seamless INA226 Revertibility: Plug-in I2C header J8 wired; firmware automatically prioritizes INA226 over I2C once installed.
- Relay Contactor Control: Two-channel 5 V relay module switching negative return lines (P-); hardware pull-ups ensure open fail-safe.
- Audio/Visual Alarm: Piezo buzzer and status LED on GPIO27 (pulsing at 1 Hz for Warning; steady on for latched Trip).
- Status & Work Done: Board B perfboard assembly & wiring actively in progress; firmware tested and verified live on ESP32.

Live ESP32 Web Dashboard & Telemetry



- Embedded Web Server: ESP32 broadcasts Wi-Fi Access Point ('PredictiveBMS') and serves live HTML5 responsive dashboard.
- Fail-Safe Safety Banner: Red top status bar explicitly indicates: 'CURRENT / VOLTAGE IS NOT SENSED · BATTERY IS DISCONNECTED'.
- Relay Contactor Status: Visual contactor indicators render in OPEN state, guaranteeing fail-safe isolation during bring-up.
- Real-Time Telemetry: 4S individual cell voltage cards, pack current, ambient/cell temperatures, and estimated thermal power.
- Air-Gapped Telemetry: HTTP server runs in low-priority FreeRTOS task, completely decoupled from the 5 Hz safety loop.

Figure: Live ESP32 web dashboard snapshot showing active Wi-Fi telemetry and fail-safe disconnect banner

Conclusion & Current Project Status

- Stage-1 Milestones Successfully Achieved:
- • Mathematical Formulation: Two-node lumped electro-thermal model and discrete Kalman observer formulated and validated.
- • Simulation Verification: KiCad 10 / ngspice circuit simulations completed; 30-scenario automated testbench passed (100% pass).
- • Predictive Advantage Proved: Acts 5.0 minutes earlier and keeps cells 27.1 °C cooler than conventional fixed-threshold BMS.
- • Embedded Firmware: Production C++17 FreeRTOS firmware built, flashed to ESP32 hardware, and executing 5 Hz safety cycle.
- • Live Web Dashboard: Operational over Wi-Fi AP with fail-safe status indicators and real-time telemetry.
- • Board A (Cell Taps): 100% constructed, soldered, calibrated, and verified on bench.
- Current Work (In Progress):
- • Board B (Supervisor): Perfboard wiring in progress with ACS712 current sensor fallback.
- Stage-2 Plan:
- • Complete Board B wiring, connect 4S1P cell pack with Layer A hardware BMS, and conduct pack thermal chamber tests.

Project Timeline

Activity / Milestone	Month 1	Month 2	Month 3	Month 4	Month 5
Milestone 1: Literature Survey & Base Paper Study	✓	✓			
Milestone 2: System Architecture & Dual-Layer Safety Design	✗	✓			
Milestone 3: Mathematical Modeling & KiCad Simulation	✗	✗	✓		
Milestone 4: Hardware Prototype: Board A & Board B Bring-Up	✗	✗	✓	In Progress	
Milestone 5: 4S Pack Testing, Enclosure & Final Viva	✗	✗			Planned

✓ = Completed
In Progress = Current Stage (Board A Done; Board B in Progress)
Planned = Stage-2 Work

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Thank You

Questions are welcome